Prática de Física dos Dispositivos Eletrônicos

FGA0100

Laboratório-4

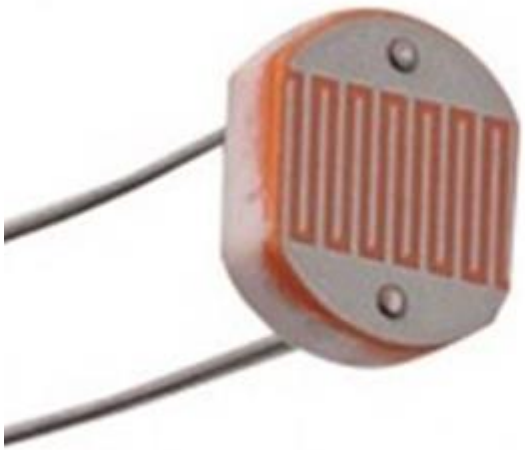
LDR

Light Dependent Resistor

01/2022

FGA **Universidade de Brasília**

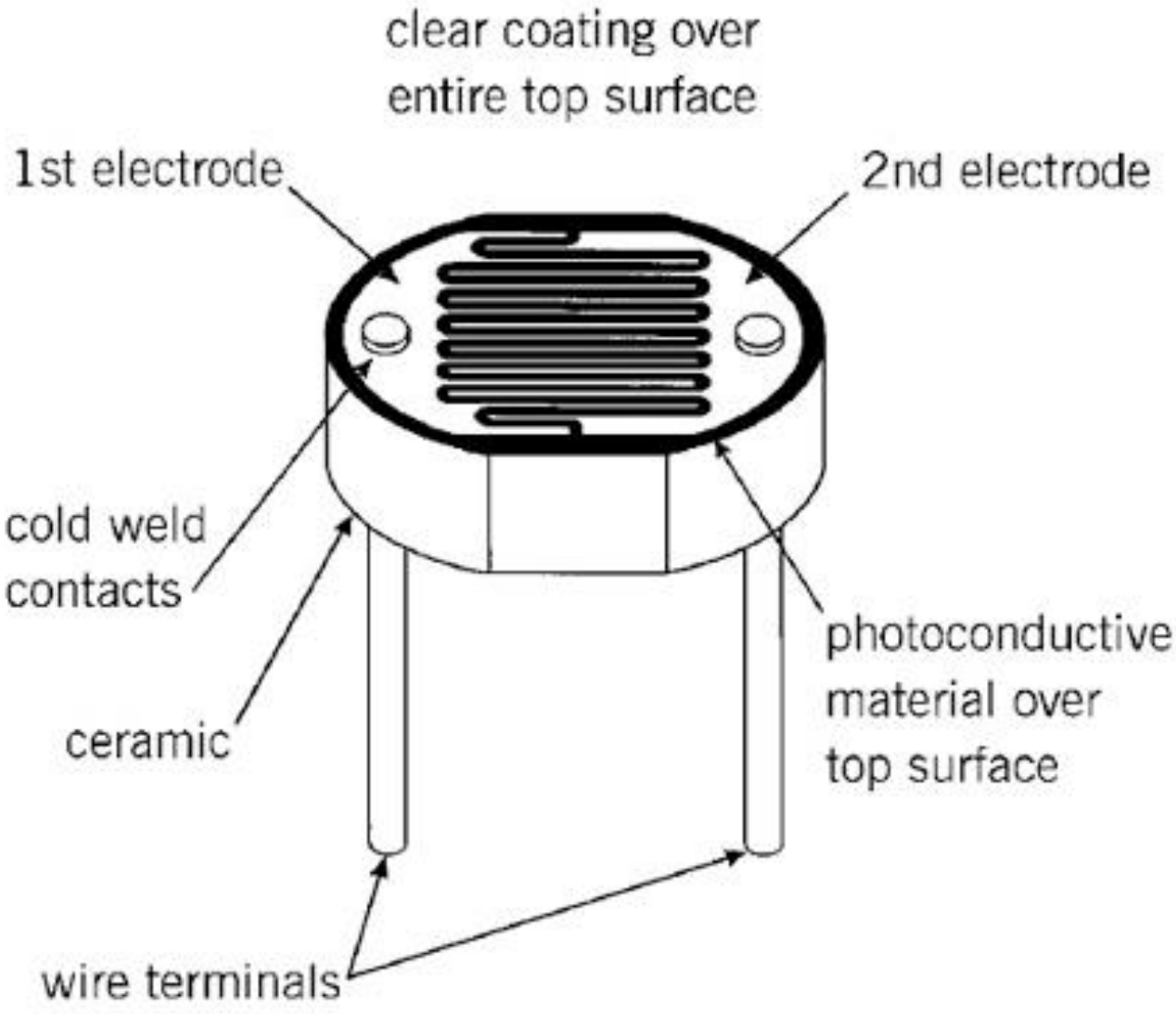
LDR: *Light Dependent Resistor*

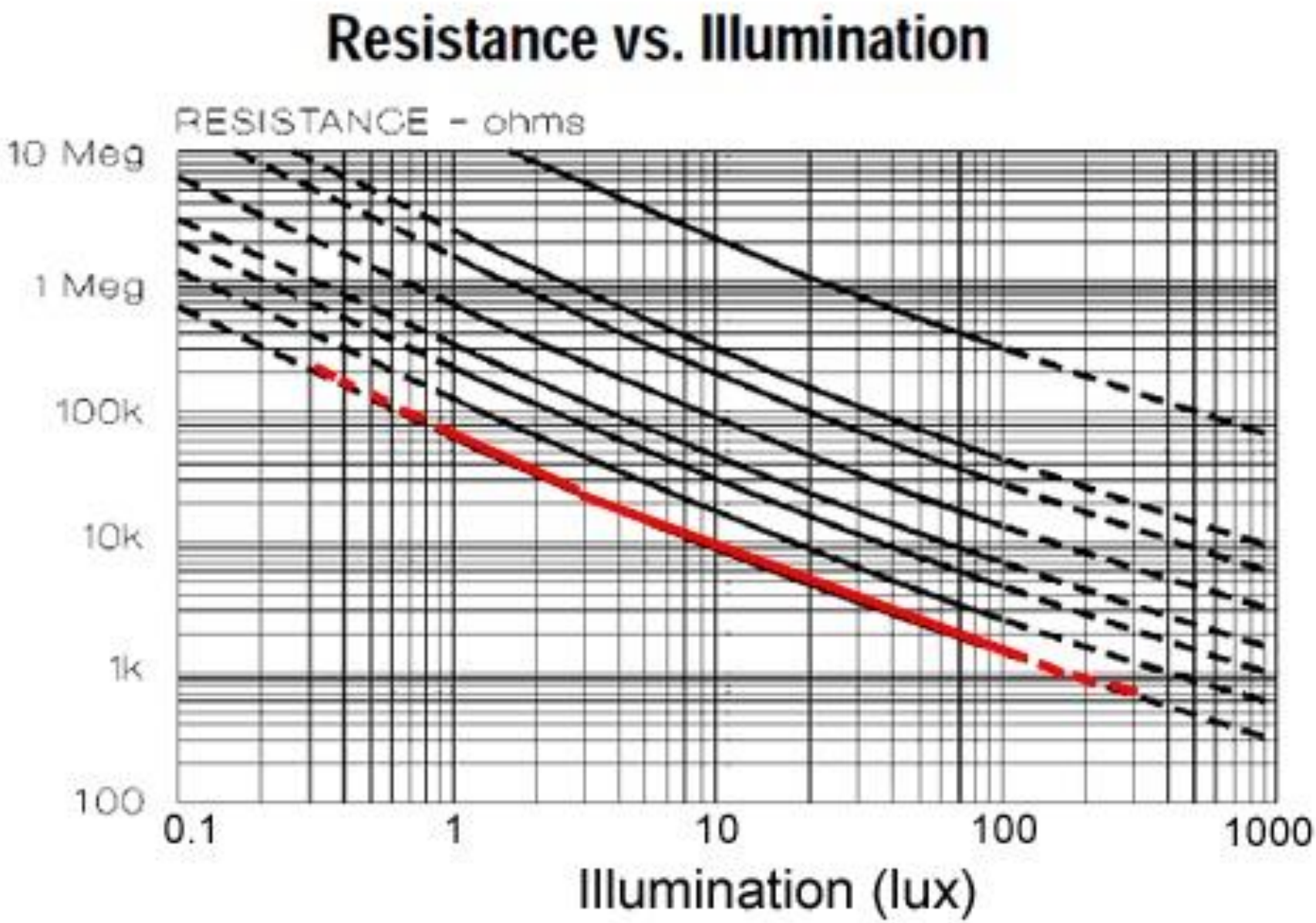


Semicondutor	Tipo	E_g, eV	
		0 K	300 K
CdS	*d*	2,582	2,42
CdSe	*d*	1,840	1,74

Restriction of Hazardous Substances Directive (RoHS): CdS e CdSe

LDR: *Light Dependent Resistor*





1 lux = 1 lúmen/m^2

Fotometria

Quantity		Unit		Dimension	Notes
Name	Symbol[nb 1]	Name	Symbol	Symbol	
Luminous energy	Q_v [nb 2]	lumen second	lm·s	**T·J** [nb 3]	Units are sometimes called *talbots*.
Luminous flux / luminous power	Φ_v [nb 2]	lumen (= cd·sr)	lm	**J** [nb 3]	Luminous energy per unit time.
Luminous intensity	I_v	candela (= lm/sr)	cd	**J** [nb 3]	Luminous power per unit solid angle.
Luminance	L_v	candela per square metre	cd/m^2	**L**$^{-2}$**·J**	Luminous power per unit solid angle per unit *projected* source area. Units are sometimes called *nits*.
Illuminance	E_v	lux (= lm/m^2)	lx	**L**$^{-2}$**·J**	Luminous power *incident* on a surface.
Luminous exitance / luminous emittance	M_v	lux	lx	**L**$^{-2}$**·J**	Luminous power *emitted* from a surface.
Luminous exposure	H_v	lux second	lx·s	**L**$^{-2}$**·T·J**	
Luminous energy density	ω_v	lumen second per cubic metre	lm·s·m^{-3}	**L**$^{-3}$**·T·J**	
Luminous efficacy	η [nb 2]	lumen per watt	lm/W	**M**$^{-1}$**·L**$^{-2}$**·T**^{3}**·J**	Ratio of luminous flux to radiant flux or power consumption, depending on context.
Luminous efficiency / luminous coefficient	V			1	
See also: SI · Photometry · Radiometry					

Fotômetro Analógico

$$P_{(\mathrm{W})} = \Phi_{\mathrm{v}(\mathrm{lm})} / \eta_{(\mathrm{lm/W})}$$

Luminous efficacy table

Light type	Typical luminous efficacy (lumens/watt)
Tungsten incandescent light bulb	12.5-17.5 lm/W
Halogen lamp	16-24 lm/W
Fluorescent lamp	45-75 lm/W
LED lamp	30-90 lm/W
Metal halide lamp	75-100 lm/W
High pressure sodium vapor lamp	85-150 lm/W
Low pressure sodium vapor lamp	100-200 lm/W
Mercury vapor lamp	35-65 lm/W

Lumens to watts table

Lumens	Incandescent light bulb watts	Fluorecent / LED watts
375 lm	25 W	6.23 W
600 lm	40 W	10 W
900 lm	60 W	15 W
1125 lm	75 W	18.75 W
1500 lm	100 W	25 W
2250 lm	150 W	37.5 W
3000 lm	200 W	50 W

Examples	
Illuminance	**Surfaces illuminated by:**
0.0001 lux	Moonless, overcast night sky (starlight)[3]
0.002 lux	Moonless clear night sky with airglow[3]
0.27–1.0 lux	Full moon on a clear night[3][4]
3.4 lux	Dark limit of civil twilight under a clear sky[5]
50 lux	Family living room lights (Australia, 1998)[6]
80 lux	Office building hallway/toilet lighting[7][8]
100 lux	Very dark overcast day[3]
320–500 lux	Office lighting[9][10][11]
400 lux	Sunrise or sunset on a clear day.
1000 lux	Overcast day;[3] typical TV studio lighting
10 000–25 000 lux	Full daylight (not direct sun)[3]
32 000–100 000 lux	Direct sunlight

Criação e Recombinação de Pares Elétron-Buraco

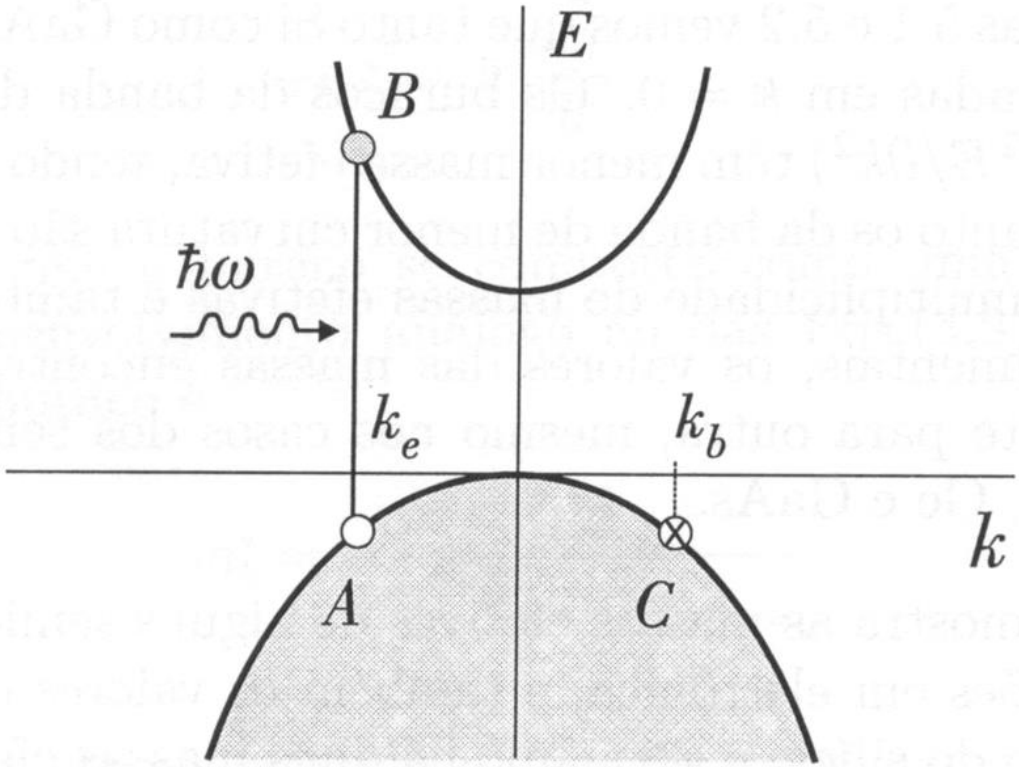


Figura 5.5: Absorção de um fóton de energia $\hbar\omega$ e vetor de onda desprezível acompanhada da criação de um par elétron-buraco em semicondutor de gap direto.

$$J = (\sigma_n + \sigma_p)\mathcal{E} = \sigma\mathcal{E}$$

$$\sigma = e(n_0\,\mu_n + p_0\,\mu_p)$$

Errado!

LDR: Resposta Espectral

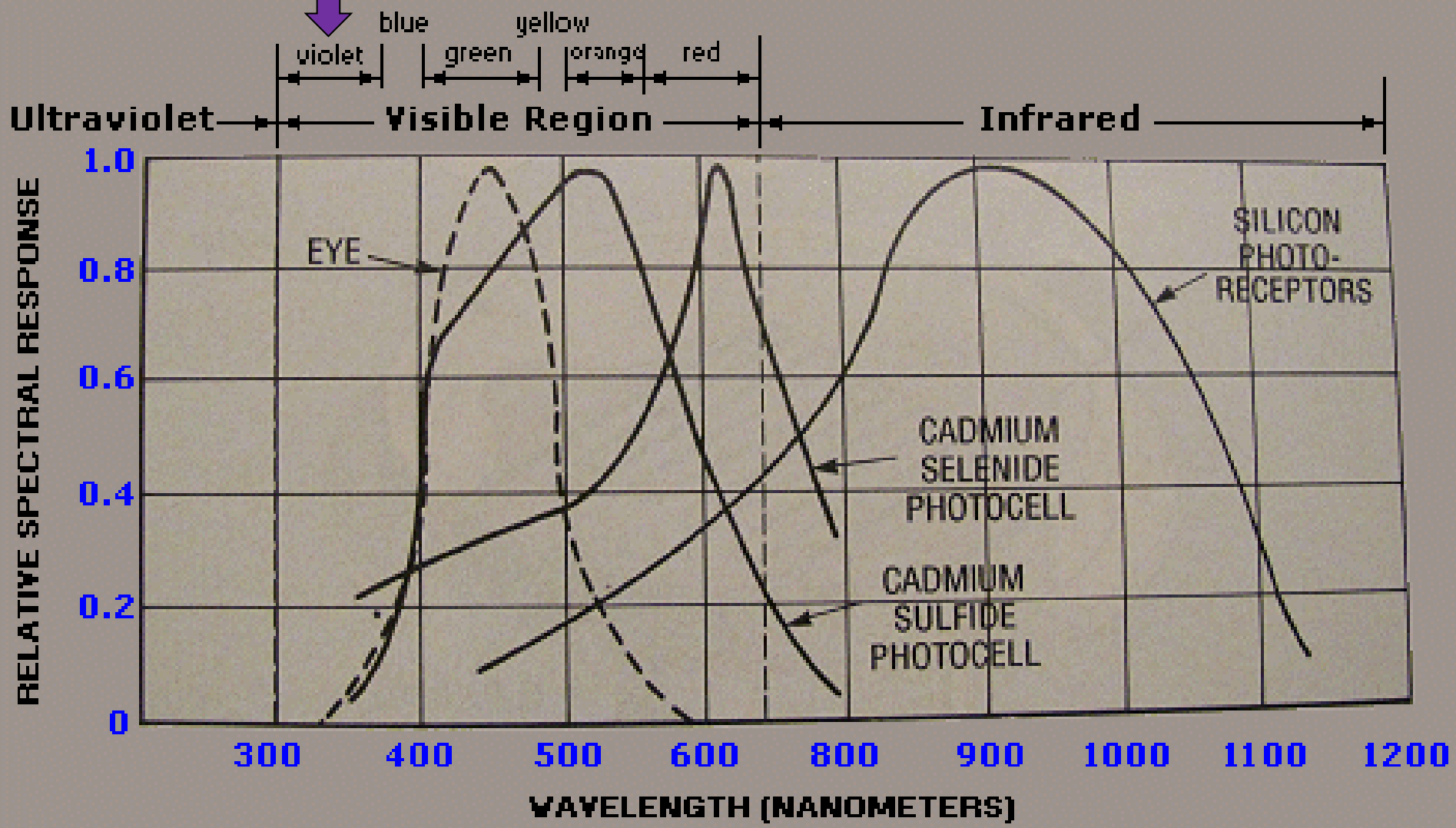
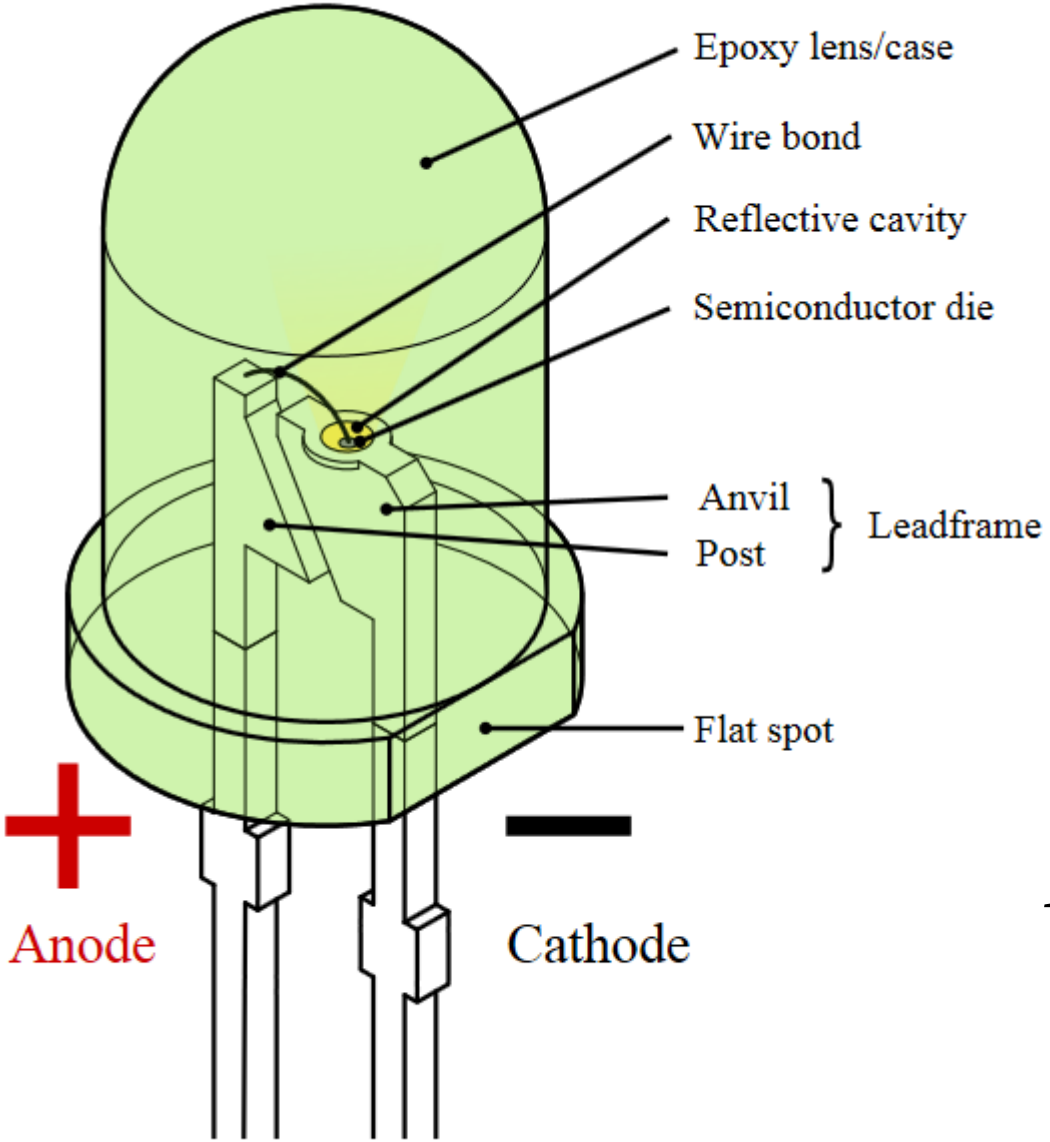


Fig. 3--Characteristic curves of photosensitive devices compared with the response curve of the human eye.

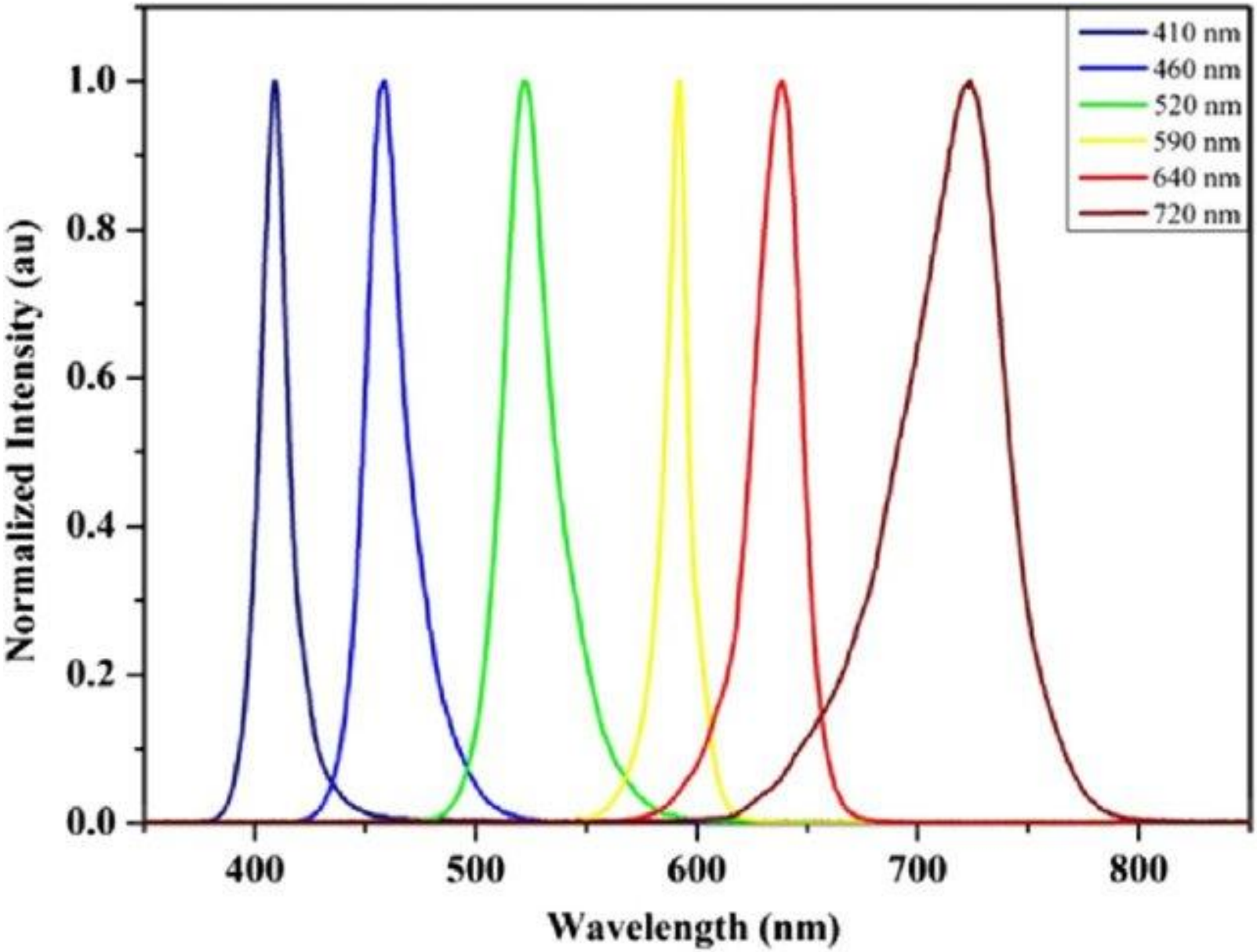
LED – Light Emitting Diode



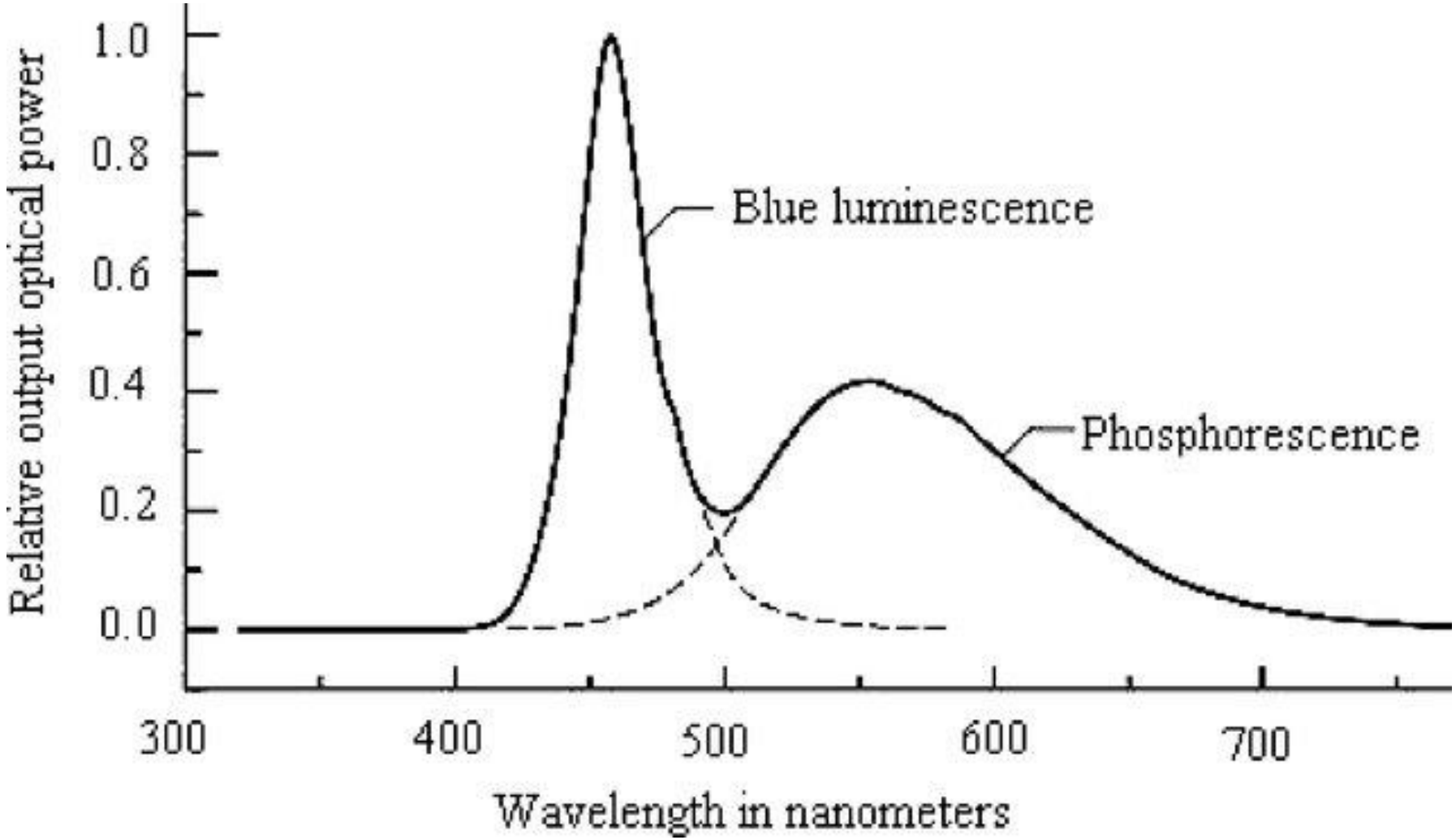
Potência Irradiada:

$$I_0 \approx k_{LED} . I_{LED}$$

LED: Espectros de Emissão



Espectro de LED Branco



Optoeletrônica - Fotoresistor

$$I(x,\lambda) = I_0 e^{-\alpha(\lambda).x} [W/m^2]$$

$$\alpha \approx 10^6 [m^{-1}]$$

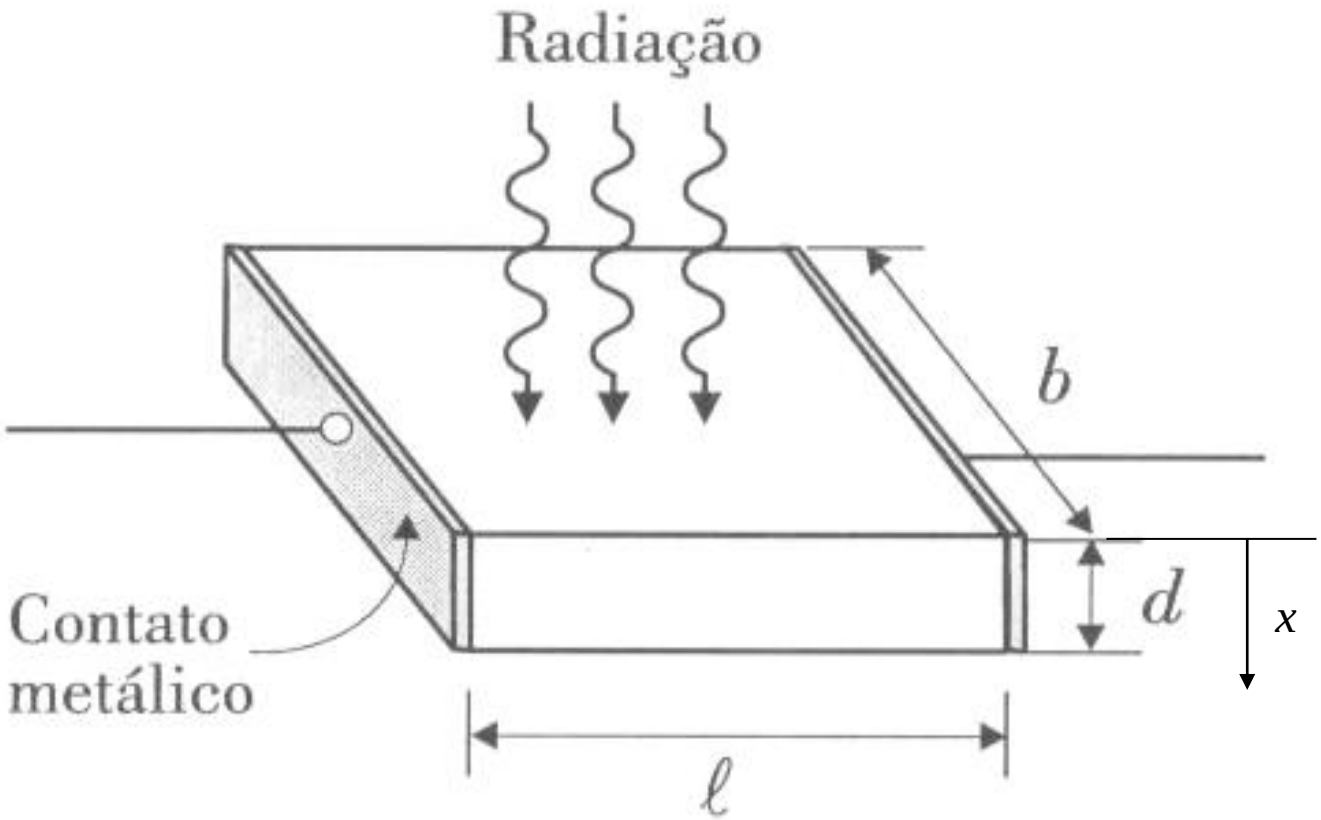


Figura 8.16: Estrutura simples de um foto-resistor, ou LDR.

Geração de Pares Elétron-Lacuna por Fótons

$$\frac{\partial \delta p}{\partial t} = \frac{\partial \delta n}{\partial t} = g = \frac{\eta I_0}{\hbar \omega d}$$

g – Taxa de Geração de pares elétron-lacuna [$m^{-3}s^{-1}$]

d – Espessura do Semicondutor (Profundidade de Absorção) [m]

I_0 – Radiação Incidente com ABSORÇÃO UNIFORME [W/m^2]

η – Eficiência de geração de pares elétron-lacuna

ω – Frequência Média dos Fótons [rad/s]

Semicondutor Intrínseco: $p_i \approx n_i$

$$\begin{cases} p = p_i + \delta p & [m^{-3}] \\ n = n_i + \delta n & [m^{-3}] \end{cases}$$

Geração e Recombinação de Pares Elétron-Lacuna

$$\begin{cases} \dfrac{\partial \delta n}{\partial t} = \cancel{g_T} + g_L - r = \dfrac{\eta I_0}{\hbar \omega d} - k_r p \delta n \\ \dfrac{\partial \delta p}{\partial t} = \cancel{g_T} + g_L - r = \dfrac{\eta I_0}{\hbar \omega d} - k_r n \delta p \end{cases}$$

$$\boxed{\tau_n = 1/k_r p} \qquad \boxed{\tau_p = 1/k_r n}$$

r – Taxa de Recombinação [1/m^3.s] k_r – Constante de Recombinação [m^3/s]

$\tau_{p,n}$ – Tempo de Recombinação [s]

Equilíbrio: $g_L = r \Rightarrow \boxed{\dfrac{\partial}{\partial t} = 0} \Rightarrow \boxed{\dfrac{\eta I_0}{\hbar \omega d} = k_r (np)}$

$$\boxed{n_i \simeq p_i} << \boxed{\delta p \approx \delta n}$$

Semicondutor Intrínseco:

$$\boxed{n \approx p}$$

$$\boxed{\delta n \approx \delta p = \sqrt{\frac{\eta I_0}{k_r \hbar \omega d}}}$$

Geração e Recombinação de Pares Elétron-Lacuna

$$\sigma = n.e.\mu_n + p.e.\mu_p = (n_i + \delta n).e.\mu_n + (p_i + \delta p).e.\mu_p$$

σ – Condutividade [S/m]

μ – Mobilidades de Elétrons e Lacunas

Semicondutor Intrínseco: $\Delta = \delta n \approx \delta p$

Condutividade: $\sigma = n.e.\mu_n + p.e.\mu_p = (n_i + \Delta)e(\mu_n + \mu_p)$

$$\boxed{\Delta = \sqrt{\frac{\eta I_0}{k_r \hbar \omega d}}}$$

LED: $\boxed{I_0 \approx k_{LED}.I_{LED}}$

I_0 – Intensidade de Fótons Emitidos [W/m^2]

$$\boxed{C = \sqrt{\frac{\eta}{k_r \hbar \omega d}}}$$

$$\boxed{\sigma = (n_i + C\sqrt{I_0})e(\mu_n + \mu_p)}$$

$$\boxed{n_i \simeq p_i}$$

Concentração Intrínseca
vs.
Temperatura em Semicondutores

$$n_i = p_i = \sqrt{n_i p_i} = (N_c N_v)^{1/2}\ \mathrm{e}^{-E_g/2k_BT}$$

$$n = N_c\ \mathrm{e}^{-(E_c - E_F)/k_BT}$$

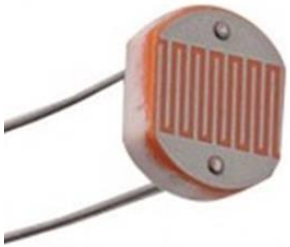
$$p = N_v\ \mathrm{e}^{-(E_F - E_v)/k_BT}$$

$$N_c = 2\left(\frac{m_c^*\ k_BT}{2\pi\ \hbar^2}\right)^{3/2}$$

$$N_v = 2\left(\frac{m_v^*\ k_BT}{2\pi\ \hbar^2}\right)^{3/2}$$

$n_i\,,\ p_i \Longrightarrow$ **Constantes**

No experimento a Temperatura ~ 300 K (INVARIANTE)
(Pequenas correntes, sem efeito Joule significativo)



Corrente vs. Tensão no LDR

$$\sigma = (n_i + C\sqrt{I_0})e(\mu_n + \mu_p)$$

$$I = G.V \qquad \frac{1}{R} = G = \frac{A\sigma}{l} \qquad I = \frac{A\sigma}{l}V$$

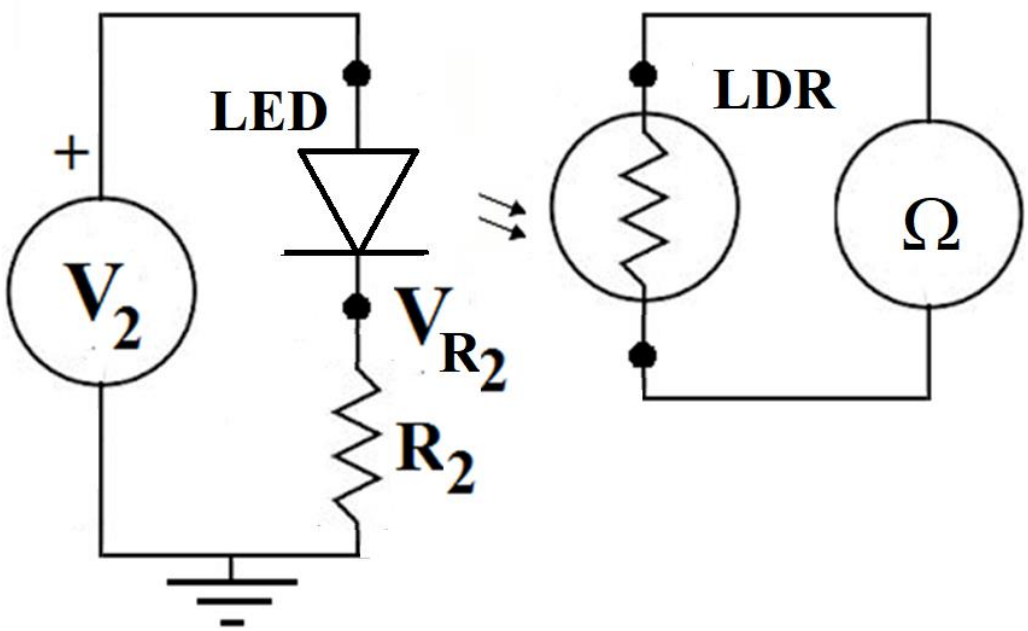
$$I = \frac{A\sigma}{l}V = \left[\frac{A}{l}(n_i + C\sqrt{I_0})e(\mu_n + \mu_p)\right]V$$

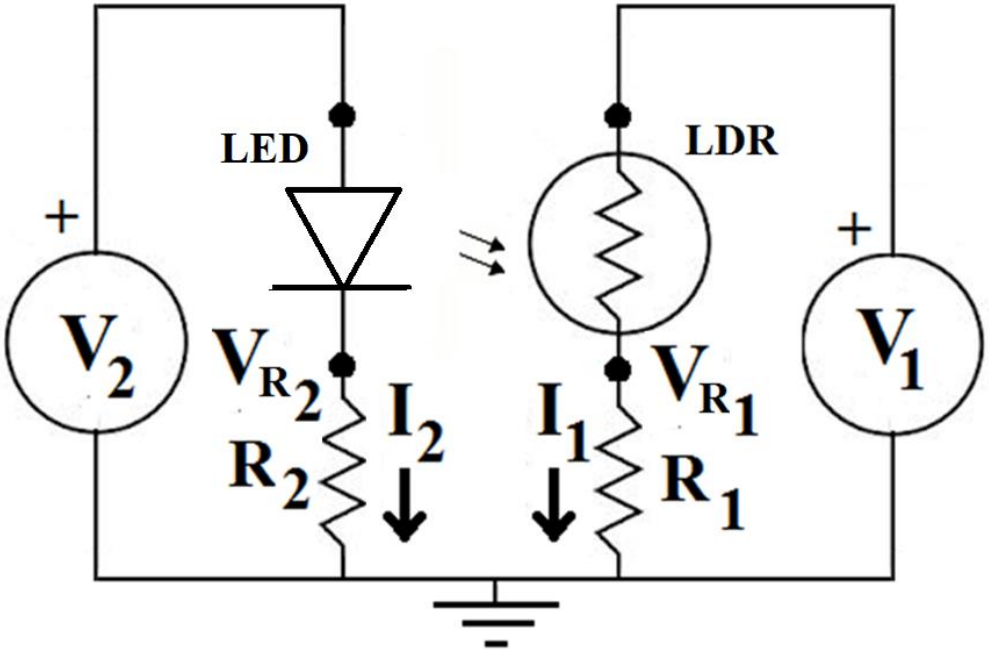
$$I = \left[\frac{A}{l}(n_i + C\sqrt{k_{LED}I_{LED}})e(\mu_n + \mu_p)\right]V$$

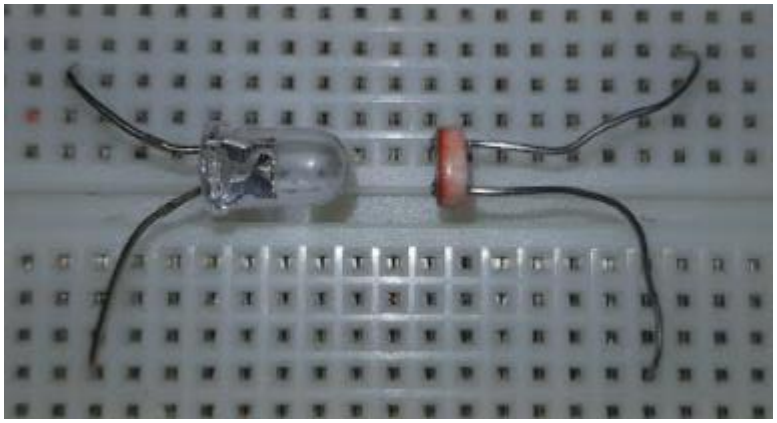
Modelo: $$\boxed{I_{LDR} = \left[C_1\sqrt{I_{LED}} + C_2\right]V_{LDR}}$$

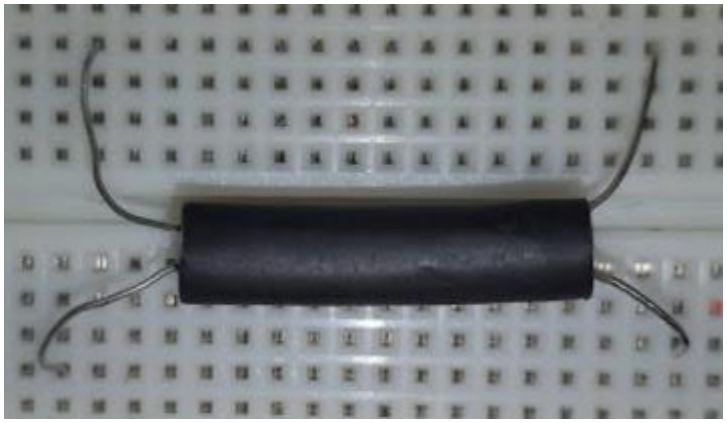
Condutância do LDR: $$G_{LDR} = C_1\sqrt{I_{LED}} + C_2$$

Montagens Experimentais

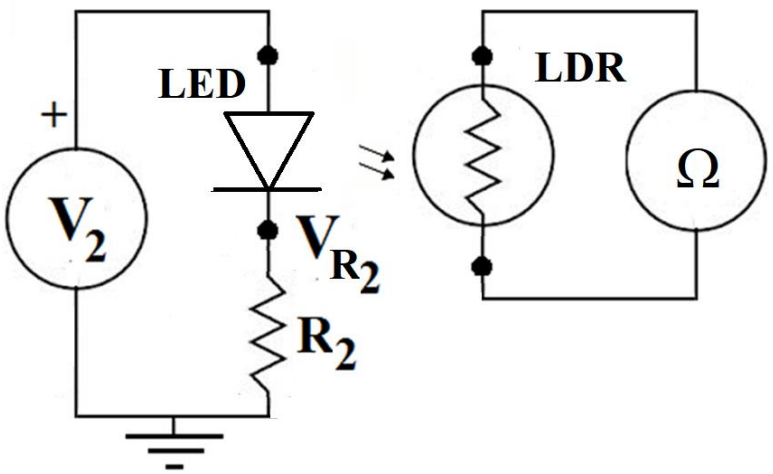








Procedimento Experimental-1



1) No circuito da **Figura-1**, faça medidas variando a tensão $\mathbf{V_2}$ e preencha a tabela com um multímetro no modo ohmímetro, medindo a resistência $\mathbf{R_{LDR}}$, e com outro multímetro no modo voltímetro, medindo a tensão $\mathbf{V_{R2}}$ sobre o resistor $\mathbf{R_2}$. Plote o gráfico $\mathbf{R_{LDR}}$ vs. $\mathbf{I_{LED}}$ (Com: $\mathbf{I_{LED}} = \mathbf{I_2} = \mathbf{V_{R2}} / \mathbf{R_2}$) ajustando a curva do modelo proposto no mesmo gráfico, usando o programa SCILAB exemplo **LED_LDR_MQ_1.sce**.

$V_{R2\ alvo}$ [V]	0,0	0,5	1,0	1,5	2,0	2,5	3,0
V_{R2} [V]							
R_{LDR} [Ω]							

Descartar

$$I_{LED} = \frac{V_{R2}}{R_2}$$

Modelo: $$G_{LDR} = C_1\sqrt{I_{LED}} + C_2$$

$$R_{LDR} = 1 / G_{LDR}$$

Exemplo SCILAB: Ajuste do Modelo

```
//Programa: LED_LDR_MQ_1.sce
//Ajuste do Modelo Não-linear do LED e LDR
clear;
//Dados Experimentais - 1
N = 6;
//Descartando ponto com Iled = 0, onde o LED está apagado.
R2 = 979; //[ohms]
VR2 = [1 2 3 4 5 6]; //[V]
Iled = VR2./R2;//[A]
Rldr = [1 2 3 4 5 6];//[ohms]
scf(1);
plot(Iled,Rldr,'or');
ylabel('Rldr [ohms]');
xlabel('Iled [A]');
title('Resistência do LDR vs. Corrente do LED');
//MODELO:
//Gldr = C1*sqrt(Iled) + C2
Gldr = 1./Rldr;
```

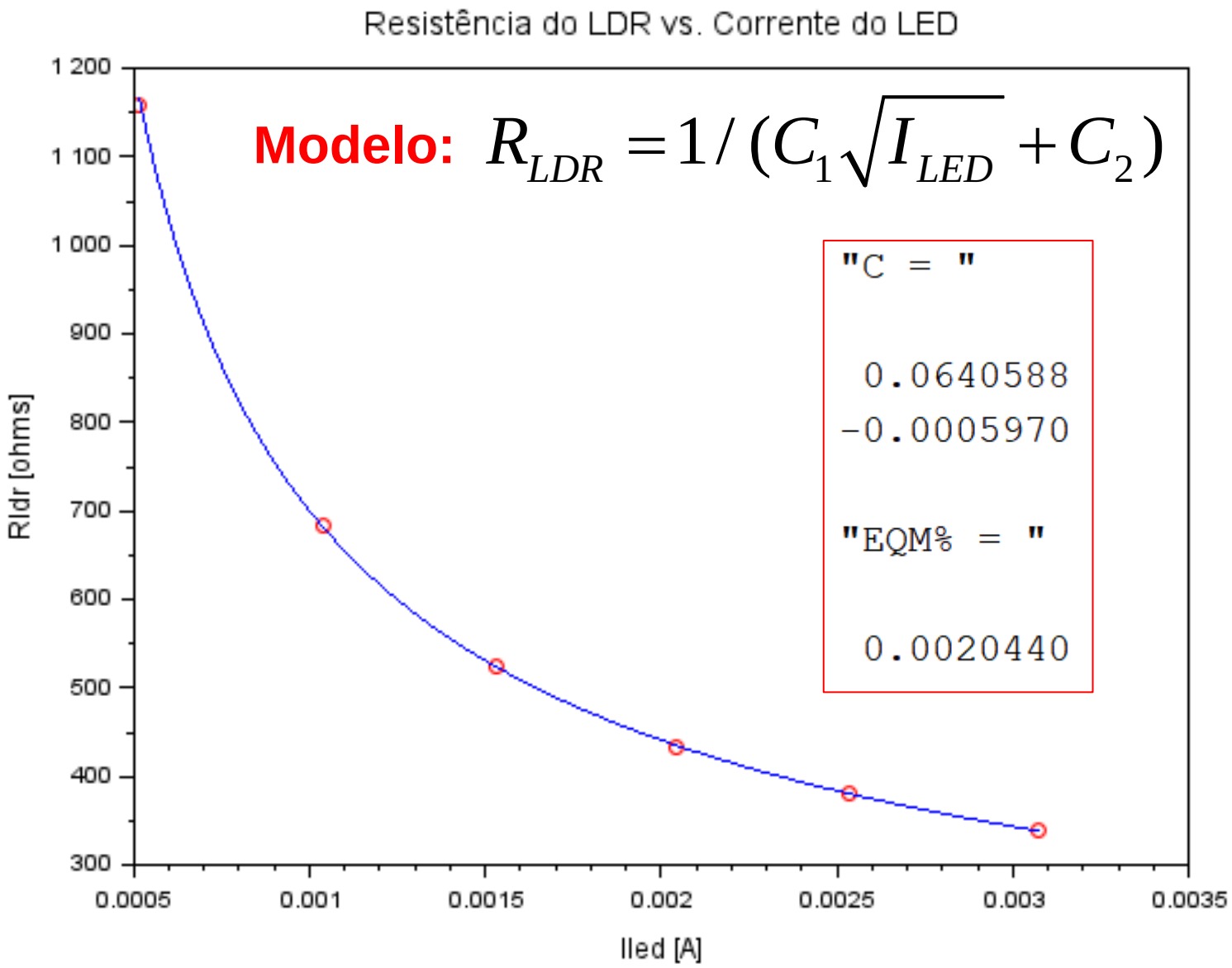
LED_LDR_MQ_1.sce

```
//Funções Base Não-Ortogonais
g1 = sqrt(Iled);
g2 = ones(1,N);
a11 = sum(g1.*g1);
a12 = sum(g1.*g2);
a21 = sum(g2.*g1);
a22 = sum(g2.*g2);
b1 = sum(Gldr.*g1);
b2 = sum(Gldr.*g2);
A = [ a11 a12; a21 a22 ];
b = [b1; b2];
C = A\b;
disp('C = ',C);
M = 100;
Iledc = linspace(min(Iled),max(Iled),M);
Gldrc = C(1)*sqrt(Iledc) + C(2);
Rldrc = 1./Gldrc;
plot(Iledc,Rldrc,'b');
Gldr0 = C(1)*sqrt(Iled) + C(2);
Rldr0 = 1./Gldr0;
//Erro Quadrático Médio Percentual
EQMP = 100*(1/N)*sum(((Rldr - Rldr0)./Rldr).^2);
disp('EQM% = ',EQMP);
```

LED_LDR_MQ_1.sce

Exemplo SCILAB: Ajuste do Modelo

LED_LDR_MQ_1.sce



Modelo: $R_{LDR} = 1/(C_1\sqrt{I_{LED}} + C_2)$

Procedimento Experimental-2

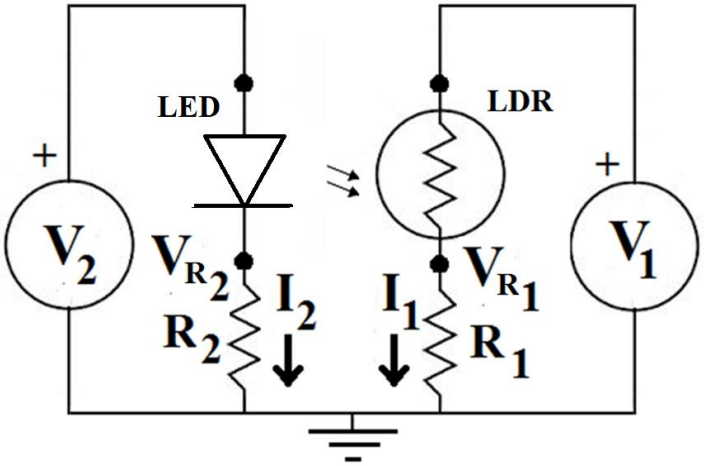
2) No circuito da **Figura-2** meça a tensão da fonte $\mathbf{V_1}$, que deve permanecer constante em 12V. Varie a tensão da fonte $\mathbf{V_2}$. Meça as tensões $\mathbf{V_{R2}}$ e $\mathbf{V_{R1}}$. Calcule os outros valores preenchendo toda a tabela, usando a quantidade correta de dígitos significativos.

$\mathbf{V_1}$ = _____+/-_____ [V]

$V_{R2\ alvo}$ [V]	V_{R2} [V]	V_{R1} [V]	I_{LED} [A]	V_{LDR} [V]	I_{LDR} [A]	G_{LDR} [S]
0,0 *						
0,5						
1,0						
1,5						
2,0						
2,5						
3,0						

$$I_{LDR} = I_1 = \frac{V_{R1}}{R_1}$$

$$G_{LDR} = \frac{I_{LDR}}{V_{LDR}}$$



$$V_{LDR} = V_1 - V_{R1}$$

$$I_{LED} = I_2 = \frac{V_{R2}}{R_2}$$

* **Descartar pontos onde o LED está apagado.**

3) Modifique o programa SCILAB exemplo **LED_LDR_MQ_1.sce** para ajustar o modelo físico-matemático da condutância $\mathbf{G_{LDR}}$ [Siemens] em função da corrente $\mathbf{I_{LED}}$ [A] pelo método dos mínimos quadrados, com os dados da tabela do **item-2**. Plote o gráfico de dispersão com a curva do modelo ajustado. Calcule o **EQMP** (Erro Quadrático Médio Percentual) e justifique se o modelo adotado é adequado. Rejeite o ponto onde o LED está apagado ($\mathbf{I_{LED}} = 0$), que não se encaixa no modelo proposto, antes de fazer o ajuste. Lembre-se que $\mathbf{G_{LDR} = 1 / R_{LDR}}$. Junto com o relatório, faça o upload do novo programa denominado **LED_LDR_MQ_2.sce**.

Modelo: $$G_{LDR} = C_1\sqrt{I_{LED}} + C_2$$

4) Questionário:

a) Pesquisa: Encontre os diferentes tipos de materiais que são usados na construção de LDRs sensíveis especificamente à luz nos comprimentos de onda das regiões do infravermelho, do visível e do ultravioleta. Cite as suas referências.

b) Pesquisa: Como a variação da temperatura afeta o comportamento do LDR? Cite as suas referências. Justifique a sua resposta.